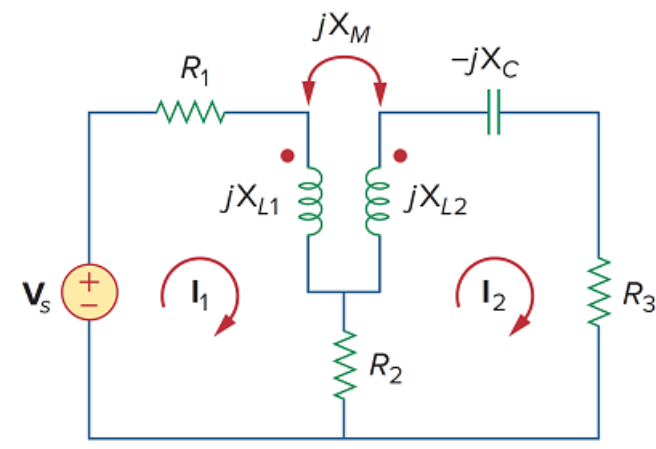


Problem

Using Fig. 13.90, design a problem to help other students better understand energy in a coupled circuit.

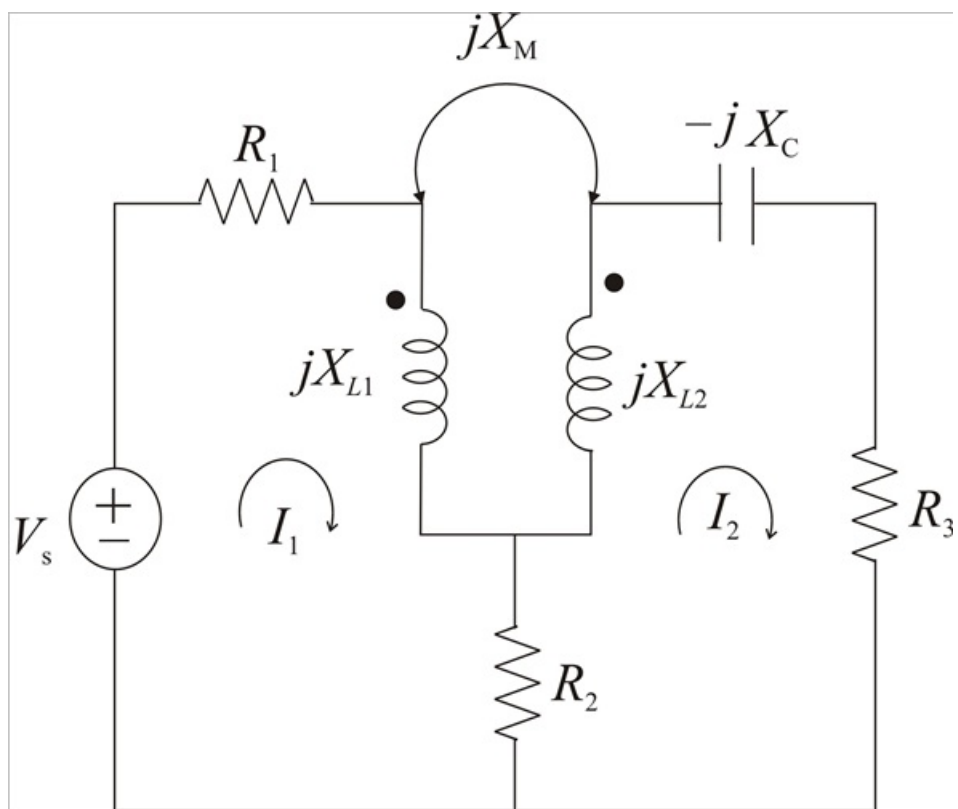
Figure 13.90



Step-by-step solution

Step 1 of 3

Given circuit is,



Step 2 of 3

Applying mesh analysis

For mesh 1,

$$-V_s + R_1 I_1 + jX_{L1} I_1 + R_2 (I_1 - I_2) - jX_M I_2 = 0$$

$$(R_1 + R_2 + jX_{L1}) I_1 - (R_2 + jX_M) I_2 = V_s \dots\dots\dots (1)$$

For mesh 2,

$$R_2 (I_2 - I_1) + jX_{L2} I_2 + (-jX_C) I_2 + R_3 I_3 - jX_M I_1 = 0$$

$$-(R_2 + jX_M) I_1 + (R_2 + R_3 + jX_{L2} - jX_C) I_2 = 0 \dots\dots\dots (2)$$

Equations (1) and (2) in matrix form as,

$$\begin{bmatrix} R_1 + R_2 + jX_{L1} & -R_2 - jX_M \\ -R_2 - jX_M & R_2 + R_3 + jX_{L2} - jX_C \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \end{bmatrix}$$

Using crammer's rule, the determinant values are

$$\Delta = \begin{vmatrix} R_1 + R_2 + jX_{L1} & -R_2 - jX_M \\ -R_2 - jX_M & R_2 + R_3 + jX_{L2} - jX_C \end{vmatrix}$$

$$\Delta = (R_1 + R_2 + jX_{L1})(R_2 + R_3 + jX_{L2} - jX_C) - (-R_2 - jX_M)^2$$

$$\Delta = \left\{ R_1 R_2 + R_1 R_3 + R_2 R_3 - X_{L1} X_{L2} + X_{L1} X_C + jR_1 (X_{L2} - X_C) \right. \\ \left. + jR_2 (X_{L1} + X_{L2} - X_C) - j2R_2 X_M + X_M^2 + jX_{L1} R_3 \right\}$$

Step 3 of 3

$$\Delta_1 = \begin{vmatrix} V_s & -R_2 - jX_M \\ 0 & R_2 + R_3 + jX_{L2} - jX_C \end{vmatrix}$$

$$\Delta_1 = V_s (R_2 + R_3 + jX_{L2} - jX_C)$$

$$\Delta_2 = \begin{vmatrix} R_1 + R_2 + jX_{L1} & V_s \\ -R_2 - jX_M & 0 \end{vmatrix}$$

$$\Delta_2 = -V_s (-R_2 - jX_M)$$

$$\Delta_2 = V_s (R_2 + jX_M)$$

Therefore,

$$I_1 = \frac{\Delta_1}{\Delta}$$

$$I_1 = \frac{V_s (R_2 + R_3 + jX_{L2} - jX_C)}{\left\{ R_1 R_2 + R_1 R_3 + R_2 R_3 - X_{L1} X_{L2} + X_{L1} X_C + jR_1 (X_{L2} - X_C) \right. \\ \left. + jR_2 (X_{L1} + X_{L2} - X_C) - j2R_2 X_M + X_M^2 + jX_{L1} R_3 \right\}}$$

And $I_2 = \frac{\Delta_2}{\Delta}$

$$I_2 = \frac{V_s (R_2 + jX_M)}{\left\{ R_1 R_2 + R_1 R_3 + R_2 R_3 - X_{L1} X_{L2} + X_{L1} X_C + jR_1 (X_{L2} - X_C) \right. \\ \left. + jR_2 (X_{L1} + X_{L2} - X_C) - j2R_2 X_M + X_M^2 + jX_{L1} R_3 \right\}}$$